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Thermodynamic assessment of the Ni–Sc binary system

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ABSTRACT

On the basis of the experimental data of the phase equilibria and the thermochemical properties, a critical evaluation for the Ni–Sc binary system has been carried out using the CALPHAD (Calculation of Phase Diagrams) method. The associated model is used for the liquid phase containing the constituent species Ni, Sc and ScNi. The terminal solid solutions Fcc_A1 (Ni), Hcp_A3 (Sc), and Bcc_A2 (Sc) are described by the solution model with the Redlich–Kister polynomial. The intermetallic compounds, $ScNi_5$, Sc_2Ni_7, ScNi and Sc_2Ni, are treated as strict stoichiometric compounds. The compound with a homogeneity range, ($ScNi_2$), is modeled using two sublattices as $(Sc,Ni)_{0.333}(Sc,Ni)_{0.667}$. A set of self-consistent thermodynamic parameters for the Ni–Sc binary system is obtained. The calculated results agree well with the available experimental data from literatures.

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1. Introduction

In recent years, the investigation of Ni-based alloys has received much attention. The intermetallic compounds containing nickel (Ni) and rare earth elements (REs) may be of special interest due to a wide range of applications as amorphous, quasicrystalline, composite and RE-substituted/doped materials. A promising $(Al_{0.84}Y_{0.09}Ni_{0.05}Co_{0.02})_{95}Sc_5$ amorphous alloy exhibits ultrahigh tensile fracture strength, high Young's modulus, good ductility and multiple shear bands [1]. The Ti–V–Ni–Sc alloys with quaternary icosahedral quasicrystals are very promising for hydrogen storage applications in Ni-metal hydride batteries owing to their good electrochemical performance [2]. The composite material composed of TiVNi-quasicrystal and Ni–RE based hydrogen storage alloy, as a composite electrode, holds high-rate discharge ability [3].The as-cast RE-substituted Zr_8Ni_{21} alloys have high capacity of gaseous hydrogen storage and fine electrochemical properties as hydrogen absorbing materials [4]. The Sc doped Ni–Co ferrites are ascribed to small crystallite sizes and large specific surface areas, and can enhance the catalytic activity for combustion of propane and acetone at moderate temperatures [5]. And nickel laterite ores are proposed as the most promising scandium resources for scandium production [6]. Therefore, the phase equilibria and the thermodynamic features of the Ni–RE systems are very helpful and useful for the development of the Ni-based alloys.

The Ni–Sc and the Ni–Yb binaries are the two systems to which the present authors are paying close attention. The Ni–Sc system has already been assessed using the CALPHAD method by Kardellass et al. [7] and their results are in fairly good agreement with most of the available experimental data on the phase diagram and thermochemical properties. However, it is possible to further achieve a better fit to experiment data on the invariant reactions and the specific heat capacity of the compound ($ScNi_2$). The Ni–Yb system has not been previously assessed thermodynamically, since the experimental information of the system is not only limited [8–13], but also the newly obtained mixing enthalpy of the liquid raises the doubtful questions to the liquidus and the melting points of some compounds [14]. The recent article [14] presented their reasonable suspicious points without further direct experimental confirmation, resulting in the thermodynamic assessment of the Ni–Yb binary system facing a difficult choice. The present work intends to assess the Ni–Sc binary system with all the updated data using a Calphad approach, and to provide a set of self-consistent thermodynamic parameters for this system. In the meanwhile, the present authors will keep in work on the investigation of the Ni–Yb system, experimentally and thermodynamically.

2. Evaluation of the available information

2.1. Literature data for the phase equilibria

With the aid of differential thermal analysis (DTA) and X-ray diffraction (XRD), Markiv et al. [15] firstly established the Ni–Sc phase diagram in the composition range <80 at% Sc and confirmed five intermediate phases, $ScNi_5$, Sc_2Ni_7, ($ScNi_2$), ScNi and Sc_2Ni. ($ScNi_2$)

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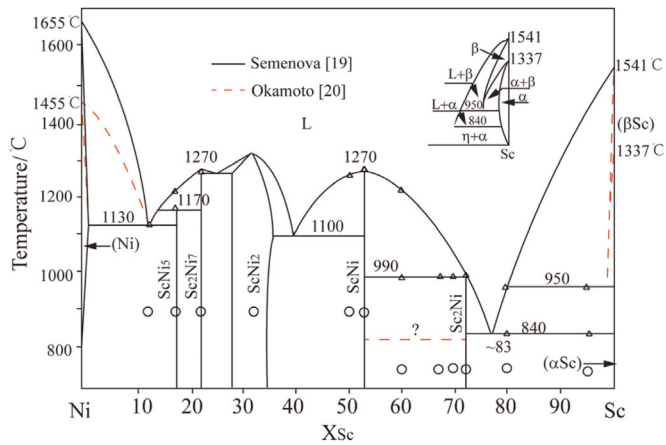

Fig. 1. The Ni–Sc binary phase diagram determined by Semenova [19] and revised by Okamoto [20].

forms a homogeneous solid solution at 1000 °C in the range of 30–36 at% Sc as reported by Markiv et al. [15], 27.2–33.4 at% Sc by Geobel and Rosen [16], and 31–35 at% Sc by Maslenkov and Braslavskaya [17], Massalski [18] revised the Ni–Sc phase diagram based on the data of Ref. [15] in the range < 80 at% Sc and the data of Ref. [17] in the range 0–36 at% Sc. Later, Semenova and Rusetskaya [19] reinvestigated the Ni–Sc system in the whole composition range 0–100 at% Sc using X-ray, metallographic, differential thermal and electron microprobe analyses, as shown in Fig. 1. There are some differences between the partial phase diagram revised in Ref. [18] and the whole phase diagram re-determined in Ref. [19]. First, the compounds ScNi and Sc_2Ni in [19] are off-stoichiometry with ~3 and ~6 at% deviation, respectively. Second, the phase diagram of the system has been proposed in the ranges > 80 at% Sc in Ref. [19] with one metatectic reaction (βSc)→L+(αSc) and one eutectic reaction L→Sc_2Ni+(αSc). Last, the eutectic reaction L→ScNi+Sc_2Ni is predicted in Ref. [18] whereas the peritectic reaction L+ScNi→ Sc_2Ni is determined in Ref. [19]. Okamoto et al. [20] reviewed the phase diagram of the Ni–Sc system, mainly based on Ref. [19] with some modifications as shown in Fig. 1. It was commented in Ref. [20] that the vast majority of the phase diagram determined in Ref. [19] is valid, except some regions for being caused by impurities of scandium in the experiment. The details of the invariant reactions in this system are shown in Table 1.

Table 1
The invariant reactions in the Ni–Sc system.

Invariant reaction	T (K)	Reaction type	x_{Sc}^{Liq}	Experiment methods	Reference
Liq↔Fcc_A1+$ScNi_5$	1403	Eutectic	0.125	DTA	[17]
	1403	Eutectic		XRD/MA/DTA/EMA	[19]
	1403	Eutectic		Revised	[20]
	1401	Eutectic	0.114	Calculated	[7]
	1411	Eutectic	0.113	Calculated	[7]
	1401	Eutectic	0.115	Calculated	This work
Liq+Sc_2Ni_7↔$ScNi_5$	1453	Peritectic	0.142	DTA	[17]
	1443	Peritectic		XRD/MA/DTA/EMA	[19]
	1443	Peritectic		Revised	[20]
	1450	Peritectic	0.147	Calculated	[7]
	1471	Peritectic	0.160	Calculated	[7]
	1443	Peritectic	0.150	Calculated	This work
Liq↔Sc_2Ni_7+($ScNi_2$)	1543	Eutectic	0.250	DTA	[17]
	1528	Eutectic		XRD/MA/DTA/EMA	[19]
	1528	Eutectic		Revised	[20]
	1548	Eutectic	0.254	Calculated	[7]
	1534	Eutectic	0.244	Calculated	[7]
	1512	Eutectic	0.257	Calculated	This work
Liq↔($ScNi_2$)+ScNi	1393	Eutectic		DTA	[17]
	1373	Eutectic		XRD/MA/DTA/EMA	[19]
	1373	Eutectic		Revised	[20]
	1427	Eutectic	0.426	Calculated	[7]
	1422	Eutectic	0.420	Calculated	[7]
	1373	Eutectic	0.417	Calculated	This work
Liq+ScNi↔Sc_2Ni	1263	Peritectic		XRD/MA/DTA/EMA	[19]
	1263	Peritectic		Revised	[20]
	1247	Eutectic	0.706	Calculated	[7]
	1254	Eutectic	0.719	Calculated	[7]
	1263	Peritectic	0.723	Calculated	This work
Liq↔Sc_2Ni+Hcp_A3	1113	Eutectic	0.83	XRD/MA/DTA/EMA	[19]
	1113	Eutectic	0.83	Revised	[20]
	1167	Eutectic	0.795	Calculated	[7]
	1139	Eutectic	0.803	Calculated	[7]
	1113	Eutectic	0.807	Calculated	This work
Bcc_A2↔Liq+Hcp_A3	1223	Metatectic	0.80	XRD/MA/DTA/EMA	[19]
	1223	Metatectic	0.80	Revised	[20]
	1229	Metatectic	0.805	Calculated	[7]
	1191	Metatectic	0.812	Calculated	[7]
	1223	Metatectic	0.807	Calculated	This work

Note: DTA, differential thermal analysis; XRD, X-ray diffraction; MA, metallographic analysis; and EMA, electron microprobe analysis.

2.2. Thermochemical data

Thermochemical data of the Ni–Sc binary system were reported by several groups. For the Ni–Sc intermetallic compounds, $ScNi_5$, ($ScNi_2$), ScNi and Sc_2Ni, De Boer et al. [21] calculated their enthalpies of formation using the electronegativity and electron density parameters of the constituent elements, and Kononov et al. [22] calculated their formation Gibbs free energies at 988 K using the chrono-potentiometry data, respectively. The standard formation enthalpies of the compounds ScNi and ($ScNi_2$) at 1473 K were measured by Selhaoui and Kleppa [23] using the direct reaction calorimetry method. The mixing enthalpies of liquid alloys up to 20 at% Sc at 1870 K were determined by Esin et al. [24] using a high temperature calorimeter and predicted by Miedema et al. [25] using the Miedema model. The heat capacities of the compound ($ScNi_2$) in the temperature range from 1.8 to 295 K were determined by Ćwik et al. [26] using a Quantum Design PPMS 14 Heat Capacity System. The enthalpies of formation of all the binary intermetallic compounds in the Ni–Sc system available from literatures are collected in Table 2.

3. Thermodynamic models

The information about the stable solid phases of the Ni–Sc [20] binary system, including the prototypes, the sublattices and the used models, is listed in Table 3.

3.1. Pure elements

The stable forms of the pure elements, Ni and Sc at 298.15 K are chosen as the reference states of the system.

Table 2
The enthalpies of formation of the intermetallic compounds in the Ni–Sc system.

Itermetallic compounds	Enthalpy of formation (kJ mol^{-1})	T (K)	Experiment methods	Reference
$ScNi_5$	−29	298	Miedema model	[21]
	−17.594	298	Redich – Kister model	[7]
	−18.068	298	Kaptay model	[7]
	−20.614	298	Associate model	This work
Sc_2Ni_7	−33	298	Interpolation	[7]
	−23.019	298	Redich – Kister model	[7]
	−24.319	298	Kaptay model	[7]
	−27.129	298	Associate model	This work
($ScNi_2$)	−53	298	Miedema model	[21]
	−43 ± 2.3	1473 ± 2	DSC	[23]
	−34.996	298	Redich–Kister model	[7]
	−37.038	298	Kaptay model	[7]
	−33.099	298	Associate model	This work
ScNi	−57	298	Miedema model	[21]
	−44.7 ± 2.3	1473 ± 2	DSC	[23]
	−46.525	298	Redich–Kister model	[7]
	−45.802	298	Kaptay model	[7]
	−41.992	298	Associate model	This work
Sc_2Ni	−20		TA	[19]
	−41	298	Miedema model	[21]
	−26.817	298	Redich–Kister model	[7]
	−25.888	298	Kaptay model	[7]
	−24.976	298	Associate model	This work

Note: DSC, direct synthesis calorimetry; TA, thermal analysis.

Table 3
Stable solid phases and the used models in the Ni–Sc binary system.

System	Phase	Strukturbericht designation	Prototype	Sublattice(s)	Used models
Ni–Sc	(Ni)[a]	A1	Cu	(Ni%,Sc)[b]	SM[c]
	$ScNi_5$	$D2_d$	$CaCu_5$	$(Sc)_{0.167}(Ni)_{0.833}$	SCM[d]
	Sc_2Ni_7		Ce_2Ni_7	$(Sc)_{0.222}(Ni)_{0.778}$	SCM
	($ScNi_2$)	C15	$MgCu_2$	$(Ni, Sc)_{0.333}(Sc, Ni)_{0.667}$	SSM[e]
	ScNi	B2	CsCl	$(Sc)_{0.53}(Ni)_{0.47}$	SCM
	Sc_2Ni		Ti_2Ni	$(Sc)_{0.667}(Ni)_{0.333}$	SCM
	(αSc)	A2	W	(Ni,Sc%)	SM
	(βSc)	A3	Mg	(Ni,Sc%)	SM

[a] The phases with homogeneity ranges are indicated with the round brackets as (phase name).
[b] The component with a larger amount is denoted by the symbol "%".
[c] SM, Solution Model.
[d] SCM, Stoichiometric Compound Model.
[e] SSM, Sublattice Solution Model.

For the pure element i in its stable or metastable state of the phase φ, the Gibbs free energy is described by an equation of the following form:

$$\begin{aligned} {}^0G_i^{\varphi}(T) &= G_i^{\varphi}(T) - H_i^{SER}(298.15\ \mathrm{K}) \\ &= a + bT + cT\ln(T) + dT^2 + eT^{-1} + fT^3 + gT^7 + hT^{-9} \end{aligned} \quad (1)$$

where $H_i^{SER}(298.15\ \mathrm{K})$ is the molar enthalpy of the element i at 298.15 K in its standard element reference (SER) state, Hcp_A3 for Sc and Fcc_A1 for Ni; and a–h are the coefficients. The molar Gibbs free energy of the element i, in its SER state, is denoted by *GHSERi* as follows:

$$GHSER_{Ni} = {}^0G_{Ni}^{Fcc_A1}(T) = G_{Ni}^{Fcc_A1}(T) - H_{Ni}^{Fcc_A1}(298.15\ \mathrm{K}) \quad (2)$$

$$GHSER_{Sc} = {}^0G_{Sc}^{Hcp_A3}(T) = G_{Sc}^{Hcp_A3}(T) - H_{Sc}^{Hcp_A3}(298.15\ \mathrm{K}) \quad (3)$$

The molar Gibbs free energy functions of pure elements are taken from the SGTE (Scientific Group Thermodata Europe) compilation by Dinsdale [27].

3.2. Solution phases

3.2.1. Terminal solid solution phases

The molar Gibbs free energies of Fcc_A1, Bcc_A2 and Hcp_A3 phases in the Ni–Sc binary system are described by the solution model, as follows:

$$G_m^{\varphi} = \sum_{i=Ni,\,Sc} {}^0G_i^{\varphi}x_i + RT\sum_{i=Ni,\,Sc} x_i \ln x_i + \Delta^E G_m^{\varphi} + \Delta^{mag}G_m^{\varphi} \quad (4)$$

where ${}^0G_i^{\varphi}$ is the molar Gibbs free energy of the pure element i in the state of the φ phase, which is taken from the SGTE database [27], x_i the mole fraction of the pure element i, R the gas constant, T the absolute temperature, $\Delta^E G_m^{\varphi}$ the excess molar Gibbs free energy and $\Delta^{mag}G_m^{\varphi}$ the magnetic contribution to the molar Gibbs free energy which equals to zero if there is no magnetic contribution.

The excess Gibbs free energy is written as a Redlich–Kister polynomial [28]:

$$\Delta^E G_m^{\varphi} = x_{Ni}x_{Sc}\sum_{n=0}^{m} {}^nL_{Ni,Sc}^{\varphi}(x_{Ni} - x_{Sc})^n \quad (5)$$

where ${}^nL_{Ni,Sc}^{\varphi}$ is the nth interaction parameter between the components Ni and Sc, which is the function of temperature:

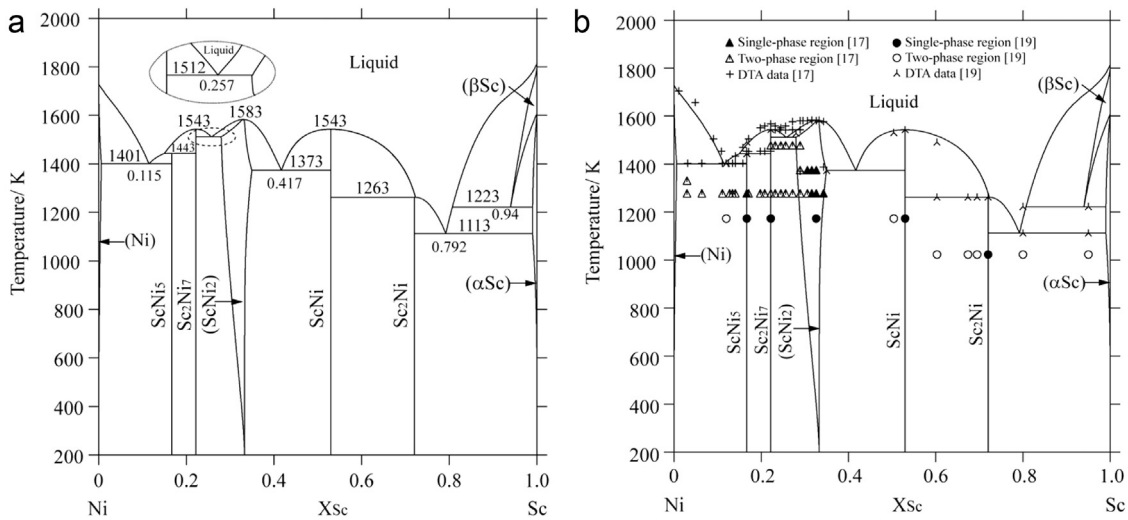

Fig. 2. The Ni–Sc binary phase diagram (a) calculated with the present description and (b) compared with the experimental data.

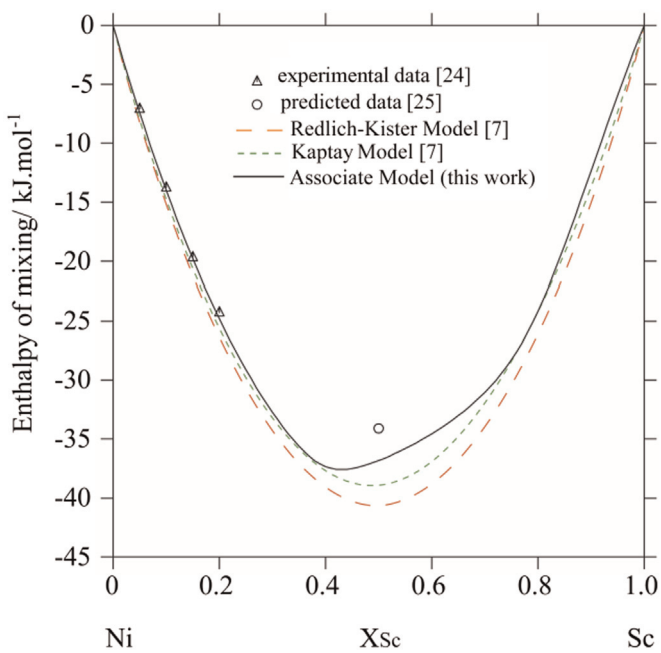

Fig. 3. The enthalpy of mixing of the liquid Ni–Sc solution at 1870 K with the reference states, Ni (liquid) and Sc (liquid).

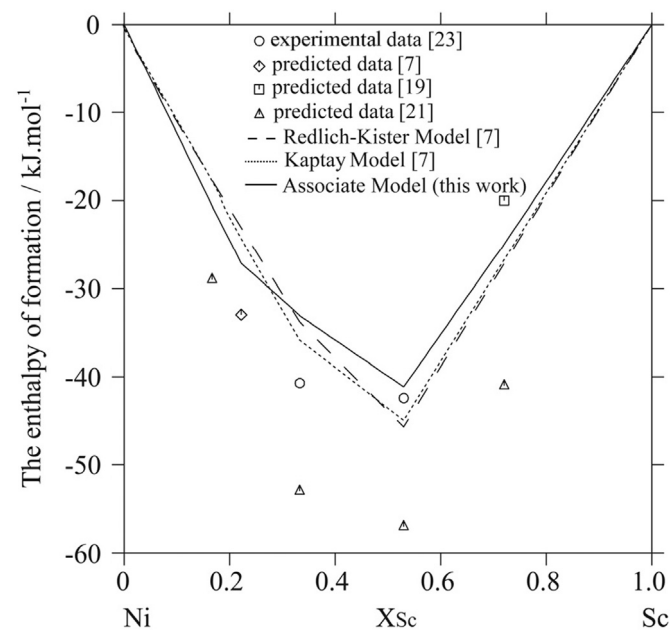

Fig. 4. The enthalpies of formation of the Ni–Sc intermetallic compounds at 298 K with the reference states, Ni (Fcc_A1) and Sc (Hcp_A3).

$$^{n}L^{\varphi}_{Ni,\,Sc} = a + bT + cT\ln(T) + dT^2 + eT^{-1} + fT^3 + gT^7 + hT^{-9} \quad (6)$$

where a–h are the coefficients to be optimized. In most cases, only the first one or two terms of the above equation are used.

The magnetic contribution is considered for the phase (αNi). It is described according to the model proposed by Hillert and Jarl [29]:

$$\Delta^{mag}G^{\varphi}_{m} = RT(\ln\beta^{\varphi} + 1)f(\tau^{\varphi}) \quad (7)$$

where β^{φ} is the Bohr magneton number, $f(\tau^{\varphi})$ a polynomial, and $\tau^{\varphi} = (T/T^{\varphi}_{c})$, and T^{φ}_{c} is the Curie temperature of solution for ferromagnetic ordering. β^{φ} and T^{φ}_{c} for the phase (αNi) are taken from the SGTE database [27].

3.2.2. Liquid phase

In order to describe reasonably the asymmetry liquid phase boundary lines and the deviated points of the invariant reactions around the stoichiometric compound ScNi in the Ni–Sc binary system, the associate model [30] is used, in consideration of the constituent species Sc, Ni and ScNi. The molar Gibbs energy of the liquid phase in the Ni–Sc binary is expressed with the following expression [28]:

$$\begin{aligned} G^{Liq}_{m} &= x_{Ni}\,{}^{0}G^{Liq}_{Ni} + x_{Sc}\,{}^{0}G^{Liq}_{Sc} + x_{ScNi}\,{}^{0}G^{Liq}_{ScNi} + RT(x_{Ni}\ln x_{Ni} \\ &\quad + x_{Sc}\ln x_{Sc} + x_{ScNi}\ln x_{ScNi}) \\ &\quad + x_{Ni}x_{Sc}\sum_{n=0}^{m}{}^{n}L^{Liq}_{Ni,Sc}(x_{Ni} - x_{Sc})^{n} + x_{Ni}x_{ScNi}\sum_{n=0}^{m}{}^{n}L^{Liq}_{Ni,ScNi}(x_{Ni} - x_{ScNi})^{n} \\ &\quad + x_{Sc}x_{ScNi}\sum_{n=0}^{m}{}^{n}L^{Liq}_{Sc,ScNi}(x_{Sc} - x_{ScNi})^{n} \end{aligned} \quad (8)$$

where ${}^{0}G^{Liq}_{i}(i = Ni,\ Sc,\ ScNi)$ and x_i are the molar Gibbs energy and the mole fraction of the pure species i in the liquid state, and ${}^{n}L^{Liq}_{i,j}$ ($n = 0,\ 1,\ 2$; $i,\ j = Ni,\ Sc,\ ScNi$, and $i \neq j$) is the nth interaction parameter between the species i and j, which is a function

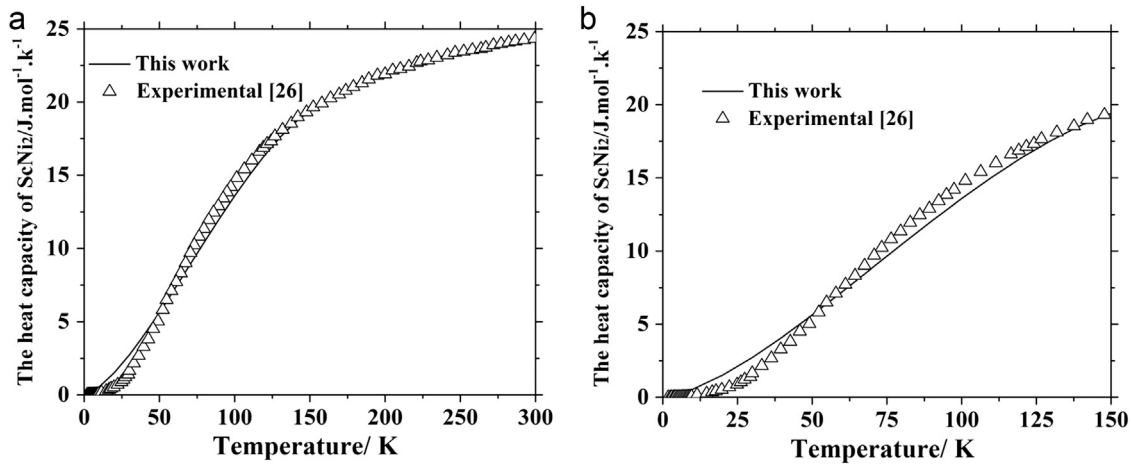

Fig. 5. The heat capacity of the compound ($ScNi_2$) (a) 0–300 K and (b) 0–150 K.

of temperature as ${}^{n}L_{i,j}^{Liq} = a + bT$ and the coefficients a and b are the parameters to be optimized in the present work.

3.3. Stoichiometric compounds

The intermetallic compounds $ScNi_5$, Sc_2Ni_7, and Sc_2Ni in the Ni–Sc binary system are stoichiometric phases. The Gibbs free energy per mole atom of Sc_pNi_q is expressed as the following:

$$G_m^{Sc_pNi_q} = \frac{p}{p+q}GHSER_{Sc} + \frac{q}{p+q}GHSER_{Ni} + \Delta G_f^{Sc_pNi_q}(T) \tag{9}$$

where $\Delta G_f^{Sc_pNi_q}(T)$ is the Gibbs free energy of formation of the stoichiometric compound Sc_pNi_q from the pure elements in their SER states. Owing to no experimental measurements available for these compounds, it is assumed that the Neumann–Kopp rule [31] applies to the heat capacity description, i.e., $\Delta C_p = 0$. Thus, $\Delta G_f^{Sc_pNi_q}(T)$ is described as

$$\Delta G_f^{Sc_pNi_q}(T) = a + bT \tag{10}$$

where the coefficients a and b are the parameters to be optimized in the present work.

3.4. Intermetallic compound solution

According to the available experimental information, the intermediate phase ($ScNi_2$) is a non-stoichiometric compound with the composition range 28–35 at% Sc which is around the strict stoichiometry 1:2. The appropriate sublattice model for this compound solution phase is adapted as $(Ni, Sc\%)_{0.333}$ $(Ni\%, Sc)_{0.667}$. The molar Gibbs free energy is described as follows:

$$\begin{aligned} G_m^{ScNi2} = {} & y_{Ni}^{I}\, y_{Ni}^{II}\, {}^{0}G_{Ni:\,Ni}^{ScNi2} + y_{Ni}^{I} y_{Sc}^{II}\, {}^{0}G_{Ni:\,Sc}^{ScNi2} + y_{Sc}^{I} y_{Ni}^{II}\, {}^{0}G_{Sc:\,Ni}^{ScNi2} + y_{Sc}^{I} \\ & y_{Sc}^{II}\, {}^{0}G_{Sc:\,Sc}^{ScNi2} + 0.333RT(\, y_{Ni}^{I} \ln y_{Ni}^{I} + y_{Sc}^{I} \ln y_{Sc}^{I}) \\ & + 0.667RT(y_{Ni}^{II} \ln y_{Ni}^{II} + y_{Sc}^{II} \ln y_{Sc}^{II}) + y_{Ni}^{I} y_{Sc}^{I} y_{Ni}^{II} \\ & L_{Ni,\,Sc:\,Ni}^{ScNi2} + y_{Ni}^{I} y_{Sc}^{I} y_{Sc}^{II} L_{Ni,\,Sc:\,Sc}^{ScNi2} + y_{Ni}^{I} y_{Ni}^{II} y_{Sc}^{II} L_{Ni:\,Ni,\,Sc}^{ScNi2} \\ & + y_{Sc}^{I} y_{Ni}^{II} y_{Sc}^{II} L_{Sc:\,Ni,\,Sc}^{ScNi2} \end{aligned} \tag{11}$$

where y_i^I and y_i^{II} are the site fractions of the component i in the first and the second sublattices, respectively. ${}^{0}G_{i:\,j}^{ScNi2}$ is the molar Gibbs free energy of the corresponding hypothetic or real compound when the first and the second sublattices are occupied by i and j respectively. $L_{i,\,j:\,k}^{ScNi2}$ and $L_{k:\,i,\,j}^{ScNi2}$ represent the interaction parameters between the elements i and j in one sublattice while the other sublattice is occupied by k (i, j, k=Ni, Sc and $i \neq j$).

The Gibbs free energy per mole atoms of the compound ${}^{0}G_{i:\,j}^{(ScNi2)}$ except ${}^{0}G_{Sc:\,Ni}^{(ScNi2)}$ is expressed according to the Neumann–Kopp rule [31] as the following:

$${}^{0}G_{i:\,j}^{(ScNi2)} = 0.333GHSER_i + 0.667GHSER_j + a + bT \tag{12}$$

For the compound ($ScNi_2$), since the heat capacity was measured and reported by Ćwik et al. [26], ${}^{0}G_{Sc:\,Ni}^{(ScNi2)}$ can be given by the same expression as used in SGTE database [27]:

$${}^{0}G_{Sc:\,Ni}^{(ScNi2)} = a + bT + cT \ln T + dT^2 + eT^3 + fT^{-1} + gT^7 \tag{13}$$

where a–g are the parameters to be optimized in the present work.

It should be pointed out that the reported heat capacity of the compound ($ScNi_2$) is in the temperature range 5–298.15 K. A segmented function is needed to fit the experimental data reasonably, especially the data at extremely low temperature. The low-temperature heat capacity, $T < 150$ K for the compound ($ScNi_2$) in the present work, is described by the following expression [32–34]:

$$C_p^{(ScNi2)} = C_{p|\mathrm{electron}} + C_{p|\mathrm{magnon}} + C_{p|\mathrm{phonon}} = cT + dT^{3/2} + eT^3 \tag{14}$$

where the first term represents the electron contribution in metal, $C_{p|\mathrm{electron}} \propto T$; the second stands for the magnon contribution in ferromagnet, $C_{p|\mathrm{magnon}} \propto T^{3/2}$; the last is the phonon contribution in solid which is the Debye T^3 law, $C_{p|\mathrm{phonon}} \propto T^3$. The coefficients c, d and e are the parameters obtained from the data of the low-temperature heat capacity. Therefore, the molar Gibbs free energy of the phase ($ScNi_2$) in the low temperature range can be derived:

$${}^{0}G_m^{(ScNi2)} = a + bT - \frac{1}{2}cT^2 - \frac{4}{15}dT^{5/2} - \frac{1}{12}eT^4 \tag{15}$$

where a and b are the parameters to be optimized with the consideration of the continuous extension between the upper and the lower temperature ranges. The conditions below need to be met:

$$C_{p1}|_{T=Tp} = C_{p2}|_{T=Tp} \tag{16}$$

$$\left.\frac{dC_{p1}}{dT}\right|_{T=Tp} = \left.\frac{dC_{p2}}{dT}\right|_{T=Tp} \tag{17}$$

Table 4
Thermodynamic parameters of the Ni–Sc binary system.

Phase	Model	Parameters
Liquid	$(Ni, ScNi, Sc)_1$	${}^0G^{Liq}_{ScNi} = -41,884.428 + 1.1678586T + 0.53GHSER_{Sc} + 0.47GHSER_{Ni}$
		${}^0L^{Liq}_{Ni,\,ScNi} = -37,328.324 - 20.999123T$
		${}^1L^{Liq}_{Ni,\,ScNi} = -3913.6137 + 21.937184T$
		${}^0L^{Liq}_{Ni,\,ScNi} = -29,611.169 + 10.346818T$
		${}^1L^{Liq}_{Ni,\,ScNi} = +9109.5345 - 0.31082938T$
		${}^2L^{Liq}_{Ni,\,ScNi} = +11,000 + 0.20128319T$
Fcc_A1	$(Ni, Sc)_1(Va)_1$	${}^0T^{Fcc_A1}_{c,\,Ni:\,Va} = 633,\ {}^0\beta^{Fcc_A1}_{Ni:\,Va} = 0.52$
		${}^0L^{Fcc_A1}_{Ni,\,Sc:\,Va} = -113,083.28 - 7.4479178T$
Bcc_A2	$(Ni, Sc)_1(Va)_3$	${}^0T^{Bcc_A2}_{c,\,Ni:\,Va} = 575,\ {}^0\beta^{Bcc_A2}_{Ni:\,Va}{}_0 = 0.85$
		${}^0L^{Bcc_A2}_{Ni,\,Sc:\,Va}{}_0 = -96,609.240 - 18.291157T$
		${}^1L^{Bcc_A2}_{Ni,\,Sc:\,Va}{}_1 = -10,866.503 - 10.636766T$
Hcp_A3	$(Ni, Sc)_1(Va)_{0.5}$	${}^0L^{Hcp_A3}_{Ni,\,Sc:\,Va}{}_0 = -50,000.231 - 16.66468T$
$ScNi_5$	$(Sc)_{0.167}(Ni)_{0.833}$	${}^0G^{ScNi5}_{Sc:\,Ni} = -22,146.431 - 5.2392832T + 0.167GHSER_{Sc} + 0.833GHSER_{Ni}$
Sc_2Ni_7	$(Sc)_{0.222}(Ni)_{0.778}$	${}^0G^{Sc2Ni7}_{Sc:\,Ni} = -28,560.089 - 7.01T + 0.222GHSER_{Sc} + 0.778GHSER_{Ni}$
$(ScNi_2)$	$(Ni,Sc)_{0.333}\ (Ni,Sc)_{0.667}$	${}^0G^{(ScNi2)}_{Ni:\,Ni} = +5000 + GHSER_{Ni}$
		${}^0G^{(ScNi2)}_{Ni:\,Sc} \begin{cases} = +11,751 + 16.346T + 4.16007\times10^{-3}T^{5/2}\ln T \\ \quad -3.0672\times10^{-7}T^4 + 10,000 + 0.667GHSER_{Ni} \\ \quad +0.333GHSER_{Sc}\ (1\,K < T < 150\,K) \\ = +41,000.902 + 120.64781T + 24.4269473\ln T \\ \quad +1.96098612\times10^{-3}T^2 - 6.48366583\times10^4T^{-1} \\ \quad +1.72045893\times10^{-3}T^3 - 9.0771641\times10^{-22}T^7 \\ \quad +8.49084458\times10^{18}T^{-9} + 10,000 + 0.667GHSER_{Ni} \\ \quad +0.333GHSER_{Sc}\ (150\,K < T < 2000\,K) \end{cases}$
		${}^0G^{(ScNi2)}_{Sc:\,Ni} \begin{cases} = -11,751 - 16.346T - 4.16007\times10^{-3}T^{5/2}\ln T \\ \quad +3.0672\times10^{-7}T^4(1\,K < T < 150\ K) \\ = -41,000.902 - 120.64781T - 24.4269473\ \ln\ T \\ \quad -1.96098612\times10^{-3}T^2 + 6.48366583\times10^4T^{-1} \\ \quad -1.72045893\times10^{-3}T^3 + 9.0771641\times10^{-22}T^7 \\ \quad -8.49084458\times10^{18}T^{-9}(150\,K < T < 2000\,K) \end{cases}$
		${}^0G^{(ScNi2)}_{Sc:\,Sc} = +5000 + GHSER_{Sc}$
		${}^0G^{(ScNi2)}_{Ni,Sc:Ni} = {}^0L^{(ScNi2)}_{Ni,Sc:Sc} = -17,800.928$
		${}^0L^{(ScNi2)}_{Ni:Ni,Sc} = {}^0L^{(ScNi2)}_{Sc:Ni,Sc} = -17,500.293$
ScNi	$(Sc)_{0.53}(Ni)_{0.47}$	${}^0G^{ScNi}_{Sc:\,Ni} = -42,000.002 - 3.0851003T + 0.53GHSER_{Sc} + 0.47GHSER_{Ni}$
Sc_2Ni	$(Sc)_{0.72}(Ni)_{0.28}$	${}^0G^{Sc2Ni}_{Sc:\,Ni} = -25,490.48 - 4.1580239T + 0.72GHSER_{Sc} + 0.28GHSER_{Ni}$

$$G_{m1}|_{T=Tp} = G_{m2}|_{T=Tp} \tag{18}$$

$$\left.\frac{dG_{m1}}{dT}\right|_{T=Tp} = \left.\frac{dG_{m2}}{dT}\right|_{T=Tp} \tag{19}$$

where C_{p1}/G_{m1} and C_{p2}/G_{m2} are the heat capacities/molar Gibbs free energies in the upper and the lower temperature ranges, respectively; Tp is the intersection point of the two related functions in the upper and the lower temperature ranges.

4. Results and discussions

Based on the lattice stabilities of all pure elements cited from Dinsdale [27] and the experimental data available from literatures, the phase relations and the thermochemical properties of the Ni–Sc binary system were carefully assessed using the PARROT module of the Thermo-Calc software [35,36]. It works by minimizing an error of sum with each piece of selected information given a certain weight. The weight was given and adjusted based upon the original

data uncertainties and upon the personal evaluation.

The parameters were optimized step by step. Based on the experimental thermochemical property and phase equilibrium data, the parameters related to solution phases (the liquid and the terminal solid solution phase) were optimized firstly. And then, the intermetallic compound phases were optimized according to their heat capacities and enthalpies of formation. Lastly, the interaction parameters of the non-stoichiometric compound phases were optimized to match their equilibrium relations.

Table 4 lists the optimized thermodynamic parameters for the Ni–Sc binary system. Using these parameters, the calculations can well reproduce most of the thermochemical property and phase boundary data. For checking of the results, the phase diagram of the Ni–Sc binary system was also calculated using Pandat software [37].

Fig. 2 shows the calculated phase diagram of the Ni–Sc system together with the experimental data taken from Refs. [10,12] and all the invariant temperatures and compositions labeled. It can be seen that the calculated phase diagram is nearly in accordance with the experimental data. The calculated incongruent melting temperatures of $ScNi_5$ and Sc_2Ni are 1443 and 1263 K, and the calculated congruent melting temperatures of Sc_2Ni_7, $ScNi_2$ and ScNi are 1543, 1583 and 1583 K respectively. The eutectic reaction temperatures for $L \leftrightarrow Fcc_A1(Ni) + ScNi_5$, $L \leftrightarrow Sc_2Ni_7 + (ScNi_2)$, $L \leftrightarrow (ScNi_2) + ScNi$ and $L \leftrightarrow Sc_2Ni + Hcp_A3(Sc)$ are 1401, 1512, 1373 and 1113 K, respectively. The metatectic reaction temperature for $Bcc_A2 \leftrightarrow Liq + Hcp_A3$ is 1223 K. The details of the invariant reactions in this binary system together with the data from literatures are listed in Table 1 for comparison. An agreement within 1 K between the present calculated temperatures and the experimentally determined data for the invariant reactions has been obtained, except the eutectic reaction $L \leftrightarrow Sc_2Ni_7 + (ScNi_2)$ for which the temperature difference is more than 16 K and the further approach to the experimental value will lead to increasing the melting point and worsening the solubility range of the compound ($ScNi_2$).

Fig. 3 presents the calculated mixing enthalpies of the liquid phase at 1870 K, which shows good agreement with the experimental data [24] and gives a better approach to the theoretical prediction of Ref. [25] than the results of Ref. [7] do.

Fig. 4 gives the assessed enthalpies of formation of all the intermetallic compounds at 298 K together with the experimental data [23] and the predicted ones in the work of Refs. [7,12,14], from which it can be seen that a fairly good agreement is obtained.

Fig. 5 shows the present calculated heat capacity of ($ScNi_2$) with comparison to the experimental data reported by Ćwik et al. [26]. The result agrees well with the experimental information.

5. Conclusions

In order to describe reasonably the asymmetry liquid phase boundary lines and the deviated points of the invariant reactions around the stoichiometric compound ScNi in the Ni–Sc binary system, the associate model is used and good results are achieved.

For the non-stoichiometric compound ($ScNi_2$), since there are the heat capacity data reported, a segmented function is used to fit the experimental data reasonably, especially the data at extremely low temperature. The expression for the low-temperature heat capacity includes the terms of the electron contribution in metals, the magnon contribution in ferromagnet and the phonon contribution in solid.

Based on the experimental data of the phase equilibria and the thermochemical properties available in literatures, the Ni–Sc binary system was thermodynamically assessed critically. A set of self-consistent thermodynamic parameters for this system has been obtained. For the Ni–Sc binary system, the comprehensive comparisons show that the calculated phase diagrams and the thermochemical properties agree well with the available experimental data.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2014.11.005.

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